

Watermarking Hardware IPs Using Design Parameter Driven Encrypted Dispersion Matrix With Eigen Decomposition Based Security Framework

Published in IEEE Access

Anirban Sengupta and Aditya Anshul, "Watermarking Hardware IPs Using Design Parameter Driven Encrypted Dispersion Matrix With Eigen Decomposition Based Security Framework," IEEE Access, vol. 12, pp. 47494-47507, 2024.

INTRODUCTION

- The application-specific hardware intellectual property (IP) cores perform several critical and computation-intensive tasks, such as digital data filtering, compression-decompression of digital data, execution of complex mathematical calculations, etc.
- High-level synthesis (HLS) is a crucial technology in the field of electronic design automation (EDA) that facilitates the design and development of complex integrated circuits or application-specific hardware IPs.
- Further, the security of the hardware IP cores has become a serious concern in the global design supply chain era.

INTRODUCTION

A. NOVELTIES OF THE PROPOSED APPROACH

- Authors presents an HLS based watermarking methodology using design parameter driven encrypted dispersion matrix with eigen decomposition based security framework for protecting hardware IP cores.
- This proposed approach presents a security framework that extracts the characteristics of the IP vendor selected design space parameters and the design space's characteristics in terms of IP vendor chosen resource configuration values and exploits them as unique features to act as digital evidence for securing hardware IP cores.
- This paper also presents the HLS design methodology of embedding a digital signature, generated using dispersion matrix, Eigen decomposition, and AES encryption block, during the register allocation phase.

RELATED WORK

Sr. No.	Existing Work	Technique Used	Remarks
1.	F. Koushanfar et. al., [11] (2005)	Multi-variable signature combinations for security	However, none of these approaches [11], [14] are robust as they can be compromised by an adversary by decoding auxiliary security parameters. Moreover, these approaches are not capable of hindering an attacker to perform RTL alteration.
2.	A. Sengupta et. al., [14] (2019)	Steganography offers security through embedded stego-constraints.	
3.	E. Castillo et.al.,[16] (2008)	employing AES-128 and MD-5 cryptosystems	However, [16] fails to provide robust security due to the generation of limited security constraints

RELATED WORK

- The following phases in the proposed approach have contributed to the improvement in the probability of coincidence, tamper tolerance and entropy:
 - a) Enhanced signature strength post encryption.
 - b) Involvement of statistical models viz. dispersion matrix and Eigen decomposition framework used for generating binary signature (digital template) and its corresponding security constraints.
 - c) The different p -bit key values for deciding the number of input resource configurations.
 - d) Design parameters chosen for dispersion matrix and Eigen matrix.
 - e) Size of the encryption key.

A. Threat Model

- A system-on-chip (SoC) integrator design house may be untrustworthy as an adversary present inside the SoC design house may pirate the original hardware IP (imported from an authentic IP vendor).
- Performing IP piracy for generating pirated designs for his/her benefit.
- Further, an adversary in the SoC integrator house may also attempt to claim the ownership rights of the hardware IP fraudulently.
- Therefore, the authentic IP vendor/seller must secure the hardware IP before exporting it to the SoC integrator house.

B. Motivation

- This proposed approach presents a novel approach for hardware IP cores using a 2D-design parameter-driven encrypted dispersion matrix with an eigen decomposition-based security watermark.
- Exploiting statistical tools such as dispersion matrix and eigen decomposition framework for generating watermark enables the IP vendor to create unique *irreproducible inimitable security mark* for embedding into the IP design.
- This proposed watermarking methodology mathematically extracts the characteristics of the IP vendor selected design space parameters and the design space's characteristics in terms of IP vendor chosen resource configuration values and exploits them as unique features to act as digital evidence for securing hardware IP cores.

THREAT MODEL AND OVERVIEW OF PROPOSED METHODOLOGY

C. Overview

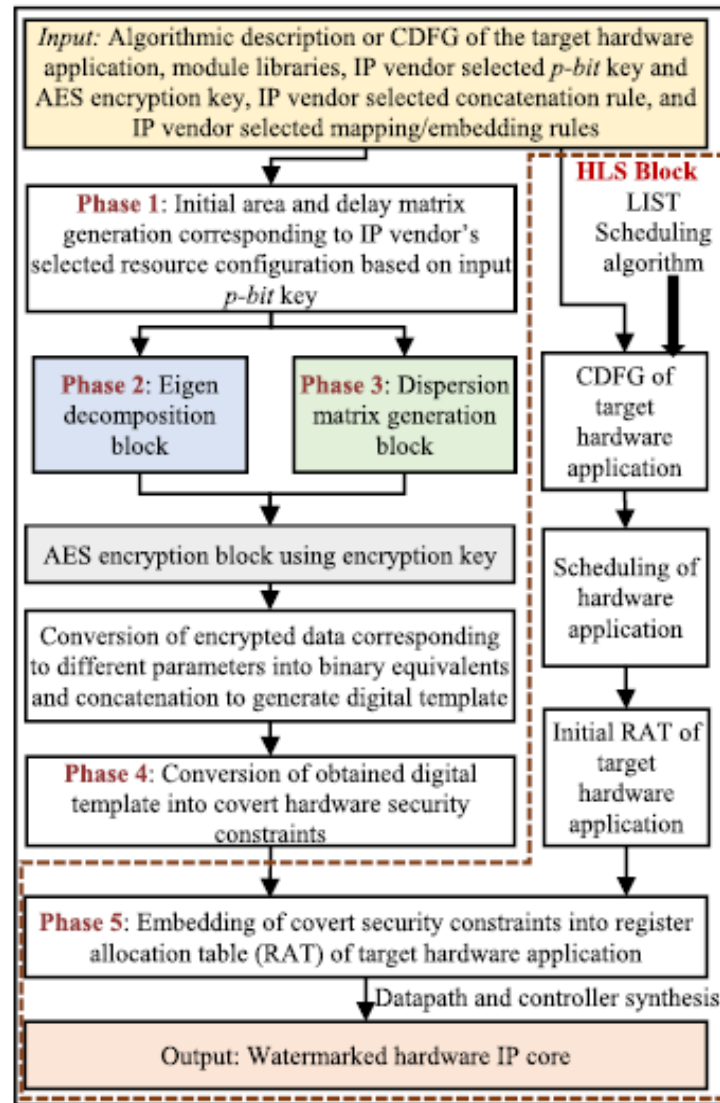


Fig. 1
Overview of the proposed security methodology.

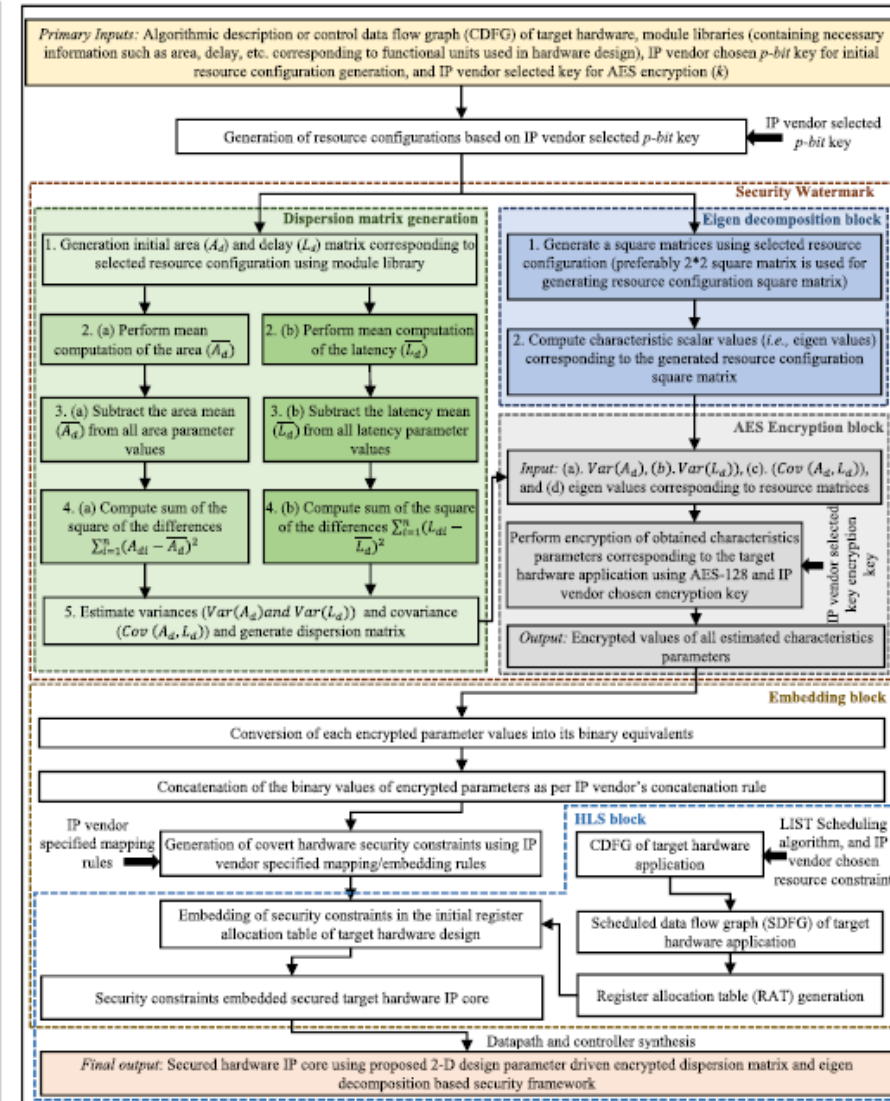
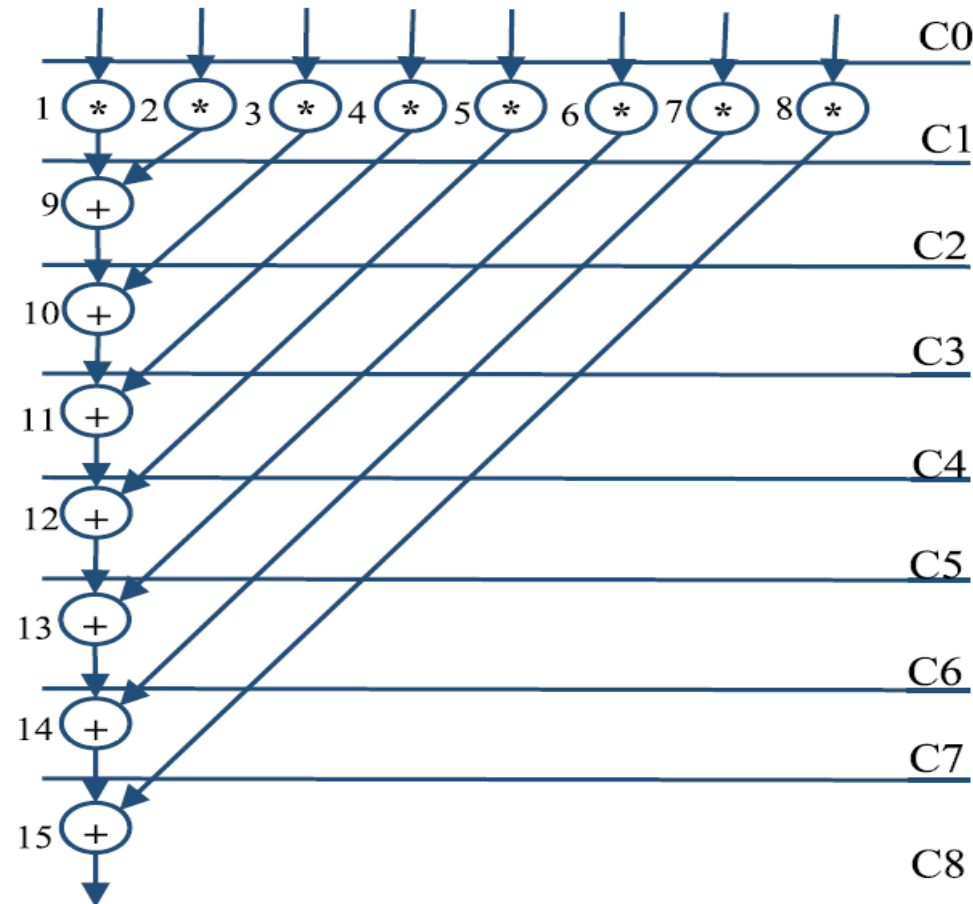


Fig. 2
Details of the proposed HLS based Watermarking methodology.

DETAILS OF THE PROPOSED METHODOLOGY

A. PHASE 1: Deriving Secret Security Data From The Encrypted Dispersion Matrix Representing The Characteristics Of The Design Space Parameter



DETAILS OF THE PROPOSED METHODOLOGY

A. PHASE 1: Deriving secret security data from the encrypted dispersion matrix representing the characteristics of the design space parameter

TABLE 1. Generation of resource configurations and its respective area (Ad) and latency (Ld) matrix corresponding to 8-point DCT based on IP vendor selected four different p-bit keys.

P-bit key	Rc (IP vendor chosen –key controlled	Ad (IP Vendor computed)	Ld (IP Vendor computed)
011	[1,4]	327 um2	927 ps
000	[1,1]	101 um2	2186 ps
111	[1,8]	629 um2	729 ps
100	[1.5]	403 um2	927 ps

PROPOSED WORK

A.1. Demonstration of generating secret security data from the encrypted dispersion matrix corresponding to the 8-point DCT application

Step 1. Computation of area (A_d) and latency (L_d) corresponding to IP vendor chosen resource configuration.

Step 2. (a). Mean computation of design parameter. $\overline{A_d} = \sum_{i=1}^n A_{di}$ (1)

Step 2. (b). Mean computation of design parameter. $\overline{L_d} = \sum_{i=1}^n L_{di}$ (2)

Step 3. (a). Subtract the mean ($\overline{A_d}$) from all area parameter values.

Step 3. (b). Subtract the mean ($\overline{L_d}$) from all latency parameter values.

Step 4. (a). Compute the sum of the square of the differences corresponding to the design area. $\sum_{i=1}^n (A_{di} - \overline{A_d})^2$ (3)

Step 4. (b). Compute the sum of the square of the differences corresponding to design latency. $\sum_{i=1}^n (L_{di} - \overline{L_d})^2$ (4)

Step 5. Estimate variance ($\text{Var}(A_d)$, $\text{Var}(L_d)$), and covariance ($\text{Cov}(A_d, L_d)$).

PROPOSED WORK

B. PHASE 2: Deriving Secret Security Data From The Encrypted Eigen Roots Representing The Characteristics Of The Design Space

The generic representation of resource configuration square matrices for extracting eigen roots are as follows:

$$A = \begin{bmatrix} P_x^1 & P_y^1 \\ P_x^2 & P_y^2 \end{bmatrix}, \quad B = \begin{bmatrix} P_x^3 & P_y^3 \\ P_x^4 & P_y^4 \end{bmatrix}$$

The generic representation of resource configuration square matrices for extracting eigen roots are as follows:

$$A = \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 8 \\ 1 & 5 \end{bmatrix}$$

PROPOSED WORK

- Subsequently, the eigen values or characteristic roots corresponding to generated matrices (i.e., A and B) are computed. The obtained eigen values corresponding to square matrices 'A' and 'B' are as follows:

$$\Rightarrow \det(\lambda I - A) = 0$$

$$\Rightarrow \det\left(\lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix}\right) = 0$$

$$\Rightarrow \det\left(\begin{bmatrix} \lambda - 1 & -4 \\ -1 & \lambda - 1 \end{bmatrix}\right) = 0$$

$$\Rightarrow \lambda^2 - 2\lambda - 3 = 0; \lambda_1 = 3 \text{ and } \lambda_2 = -1$$

Similarly, $\lambda_3 = 6.46$ and $\lambda_4 = -0.46$

- Further, as discussed in the overview subsection, the obtained characteristic roots corresponding to the IP vendor selected design space are encrypted using the AES-128 encryption mechanism.

PROPOSED WORK

- Next, all encrypted values are converted into binary equivalents to generate secret data corresponding to all design space characteristic roots. The final secret data corresponding to all four characteristic roots are:

$$\lambda_1^{NE} = 3$$

$$\lambda_1^E = 917\text{ebc}40242\text{d1c}2\text{c}362\text{b041}30\text{ae}4\text{a4a}7.$$

$$\lambda_1^E = 0010001011111010111100010000000010010000101101000111000010110000110110001010110000101011000001000001001100001010111001001010010010100111.$$

Similarly calculation for λ_2^E , λ_3^E , λ_4^E .

- The generation of the final encrypted signature and its corresponding covert hardware security constraints is explained in the next subsection.

PROPOSED WORK

C. PHASE 3: CONCATENATION OF THE ENCRYPTED SIGNATURE AND ITS RESPECTIVE ENCODING INTO HARDWARE SECURITY CONSTRAINTS

➤ The IP vendor selected concatenation rule for the proposed approach is

$$Var(Ad)E \parallel Var(Ld)E \parallel Cov(Ad, Ld)E \parallel \lambda_1^E \parallel \lambda_2^E \parallel \lambda_3^E \parallel \lambda_4^E,$$

where $\parallel$ is the concatenation operator.

Note: The concatenation order can vary as per the IP vendor's choice. The final obtained encrypted signature is: “00111011001111001011111000.
.00011100011101001 (896-bit). Subsequently, the generated encrypted signature is transformed into covert hardware security constraints using the IP vendor chosen encoding/mapping mechanism for embedding into the design.

➤ The IP vendor chosen encoding mechanism used in the proposed approach is as follows, where $C0$ and $C1$ represent the covert hardware security constraints, $\langle Lx, Ly \rangle$ denotes the storage variable pairs in the RAT of the design:

PROPOSED WORK

$$C0 = \{(L2a, L2b), a, b \in W \text{ and } (0 \leq a \leq m), (0 \leq b \leq n)\} \quad (11)$$

$$C1 = \{(L2a+1, L2b+1), a, b \in W \text{ and } (0 \leq a \leq s), (0 \leq b \leq t)\} \quad (12)$$

- For incoming bit '0', covert security constraints are generated using $C0$, otherwise using $C1$. The upper limits of ' a ' are ' m ' and ' s ', respectively, while the upper limits for ' b ' are ' n ' and ' t '. The upper limit depends on the maximum number of storage variables in the target application's SDFG.
- For example, using the above representation, the determined secret hardware security constraints for 8-point DCT can be obtained as follows: $\langle L0, L2 \rangle$, $\langle L0, L4 \rangle - \langle L0, L30 \rangle$, $\langle L2, L4 \rangle - \langle L2, L30 \rangle$, $\langle L4, L6 \rangle - \langle L4, L30 \rangle$, $\langle L6, L8 \rangle - \langle L6, L30 \rangle$, $\langle L8, L10 \rangle - \langle L20, L28 \rangle$, $\langle L1, L3 \rangle$, $\langle L1, L5 \rangle - \langle L1, L29 \rangle$, $\langle L3, L5 \rangle - \langle L3, L29 \rangle$, $\langle L5, L7 \rangle - \langle L27, L29 \rangle$.
- These generated security constraints are embedded into the design of the 8-point DCT application.

D. Generating the Hardware Security Constraints from Obfuscated Scheduled Design and Fingerprint Biometric .

For the sake of brevity, we are considering the fingerprint signature strength of 100 bits (including first 50 and last 50 bits of fingerprint signature). The first 50 bits of the generated fingerprint signature are:

“11011001-110011-1-10111-10111100-1000001-1-101100101-11000”.

PROPOSED WORK

D. PHASE 4: Embedding Of Hardware Security Constraints Into The Design To Act As Secret Digital Evidence

- Initially, the CDFG is scheduled using IP vendor resources and LIST scheduling to create an SDFG.
- An initial register allocation table (RAT) is generated and updated with security constraints, ensuring no two variables sharing constraints use the same register.
- Conflicts are resolved by adjusting registers or allocating new ones, embedding digital evidence to counter IP piracy.
- The final secured RAT is used to generate the secure RTL datapath, as illustrated in Table 2.

PROPOSED WORK

CS	Red(R)	Green (G)	Indigo (I)	Blue (BL)	Yellow (Y)	Black (B)	Violet (V)	Pink (P)	Lime (LI)	Olive (O)	Aqua (A)	Teal (T)	Grey (Gr)	Magenta (M)	Silver (S)	Khaki (K)
0	L0	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12	L13	L14	L15
1	L16/L17	L17 /L16	L2	L3	L4	L5	L6	L7	-	-	-	-	-	-	-	-
2	L24	-	L18 /L19	L19 /L18	L4	L5	L6	L7	-	L24	-	-	-	-	-	-
3	L25	-	L19	L19	L20 /L21	L21 /L20	L6	L7	L25	-	-	-	-	-	-	-
4	L26	-	-	-	L20 /L21	L21 /L20	L22 /L23	L23 /L22	-	L26	-	-	-	-	-	-
5	L27	-	-	-	L21	L21	L22 /L23	L23 /L22	-	-	L27	-	-	-	-	-
6	L28	-	-	-	-	-	L22 /L23	L23 /L22	-	L28	-	-	-	-	-	-
7	L29	-	-	-	-	-	L23	L23	-	-	-	L29	-	-	-	-
8	L30	-	-	-	-	L30	-	-	-	-	-	-	-	-	-	-

TABLE 2. Register allocation table pre and post embedding generated signature.

PROPOSED WORK

E. Validation of the secured design using the proposed approach

- The goal of this paper is to provide robust validation of secured IP designs.
- For accomplishing this, the watermark constraints are extracted from the design-under-test (DUT) chip and matched with the originally embedded watermark security constraints of the IP design.
- From the extracted layout design file of the DUT chip, through reverse engineering, the IP core register transfer level (RTL) files are obtained.
- Finally, the watermarking constraints are extracted from the IP core RTL file (hardware description language code) for matching. In case of a complete match, validation of authentic/secured designs is complete.

F. IP ownership conflict resolution process and Ip piracy detection

- Initially, the CDFG is scheduled with IP vendor resources and LIST scheduling to create an SDFG.
- An initial register allocation table (RAT) is then updated with security constraints, avoiding conflicts by adjusting registers or allocating new ones.
- This secured RAT, embedding digital evidence, helps prevent IP piracy and generates the secure RTL datapath.

A. Security analysis of the proposed approach

- The probability of coincidence is a measure of the likelihood of coincidental detection of the exact security constraints in an unsecured hardware IP core design.

$$C_i = \left(1 - \frac{1}{r}\right)^c \quad (1)$$

- *where 'r' is the total registers present in the design of the target hardware application before embedding the security constraints, and 'c' is the strength of embedded security constraints.*

RESULTS

TABLE 3. Comparison of probability of coincidence (CI) between the proposed approach, [10], [16].

Benchmark	Register count before embedding security constraints	Author's Approach		[11]		[17]	
		Embedding security constraints	(C_i)	Embedding security constraints	(C_i)	Embedding security constraints	(C_i)
8-point DCT	16	214	1.00E-06	81	5.36E-03	160	3.27E-05
FIR	16	343	2.43E-10	81	5.36E-03	160	3.27E-05
ARF	16	441	4.35E-13	81	5.36E-03	160	3.27E-05
DWT	10	164	3.13E-08	81	1.96E-04	160	4.77E-08
JPEG-CODEC	137	896	1.41E-03	81	5.52E-01	160	3.09E-01

RESULTS

TABLE 4. Comparison of probability of coincidence (CI) between the proposed approach, [11], [17].

Benchmark	Register count before embedding security constraints	Author's Approach		[11]		[17]	
		Embedding security constraints	(C_i)	Embedding security constraints	(C_i)	Embedding security constraints	(C_i)
8-point DCT	16	214	1.00E-06	128	2.58E-04	199	2.64E-06
FIR	16	343	2.43E-10	128	2.58E-04	199	2.64E-06
ARF	16	441	4.35E-13	128	2.58E-04	199	2.64E-06
DWT	10	164	3.13E-08	128	1.39E-06	199	3.13E-08
JPEG-CODEC	137	896	1.41E-03	128	3.91E-01	199	2.32E-01

RESULTS

- Tamper tolerance measures security in terms of the brute force attempt performed by an adversary to tamper the design or guess the exact signature combination. Defined as, $Z_t = K^c$ (2)

TABLE 5. Comparison of tamper tolerance (ZT) between the proposed approach, [10], [11], [16], and [17].

Benchmark	Author's Approach	[10]	[16]	[11]	[17]
8-point DCT	2.63E+64	2.41E+24	1.46E+48	3.40E+38	8.03E+59
FIR	1.79E+103	2.41E+24	1.46E+48	3.40E+38	8.03E+59
ARF	5.56E+132	2.41E+24	1.46E+48	3.40E+38	8.03E+59
DWT	2.33E+49	2.41E+24	1.46E+48	3.40E+38	8.03E+59
JPEG-CODEC	5.28E+269	2.41E+24	1.46E+48	3.40E+38	8.03E+59

RESULTS

TABLE 6. Comparison of probability of coincidence (CI) and tamper tolerance (ZT) with variation in signature strength corresponding to JPEG-CODEC IP for the proposed approach.

P-bit key based resource configuration	Variation in signature	C_i	Z_t
[1,4],[1,1] and $E_K=128$ bit	512 bit	2.34E-02	1.34E+154
[1,4],[1,1],[1,8],[1,5] and $E_K=128$ bit	896 bit	1.41E-03	5.28E+269
[1,4],[1,1],[1,8],[1,5] and $E_K=256$ bit	1792 bit	1.98E-06	2.79E+539

RESULTS

- The design cost analysis of the proposed approach is performed using equation (3) [22].

$$C_T = J_1 \left(\frac{L_T}{L_{max}} \right) + J_2 \left(\frac{A_R}{A_{max}} \right) \quad (3)$$

TABLE 7. Results of author's proposed approach pre and post implanting generated signature.

Benchmarks	IP vendor selected resource configuration for scheduling	Initial design (<i>i.e.</i> , pre signature implanted unsecured design)				Final secured signature implanted design				Design cost overhead %
		Area (μm^2)	Latency (ps)	Register count	Design cost	Area (μm^2)	Latency (ps)	Design cost	Register count	
8-point DCT	1(+), 2(*)	182.45	1324.86	16	0.446	182.45	1324.86	0.446	16	0
FIR	1(+), 2(*)	106.95	2583.46	16	0.569	109.31	2583.46	0.57	19	0.17
ARF	1(+), 2(*)	182.45	2450.98	16	0.412	187.95	2450.98	0.415	23	0.72
DWT	2(+), 3(*)	272.10	1722.31	10	0.656	275.25	1722.31	0.657	14	0.15
JPEG- CODEC	6(+), 8(*)	824.96	3245.89	137	0.157	824.96	3245.89	0.157	137	0

RESULTS

- The Entropy measures the degree of uncertainty or the amount of unpredictable randomness that an attacker must overcome to ascertain the secret signature [35]. The entropy of the proposed approach can be represented as follows.

$$E_t = 1/2^d * 1/2^k * 1/2^p * \left(\frac{1}{A_M * M_M} \right)^t \quad (4)$$

TABLE 8. Comparison of entropy between the proposed approach, [10], [11], [16], and [17].

Benchmark	Author's Approach	[10]	[16]	[11]	[17]
8-point DCT	1.09E-544	1.03E-32	2.01E-87	8.63E-78	3.08E-65
FIR	3.33E-549	1.03E-32	2.01E-87	8.63E-78	3.08E-65
ARF	1.06E-547	1.03E-32	2.01E-87	8.63E-78	3.08E-65
DWT	6.99E-547	1.03E-32	2.01E-87	8.63E-78	3.08E-65
JPEG-CODEC	9.94E-577	1.03E-32	2.01E-87	8.63E-78	3.08E-65

RESULTS

- Table 9 illustrates a comparison between the entropy of the proposed approach, cryptographic key random number generator (RNG) [11] and a semiconductor superlattice true random number generator (SSL-TRNG) [30] based approaches.
- The proposed approach demonstrates enhanced entropy, indicated by lower probability values compared to [11] and [30].

TABLE 9. Comparison of entropy between proposed approach and RNG techniques (Crypto key based) for JPEG-CODEC.

Author's Approach	Key based RNG [11]	SSL TRNG [30]
9.94E-577	8.63E-78	9.33E-302

RESULTS

- This is because the proposed approach offers more resistance and uncertainty to an attacker than other watermarking approaches [10], [11], [16], [17], and [30].
- More explicitly, an attacker needs to decode the following resistive parameters to overcome the uncertainty in regenerating or forging the exact security constraints, which offers stronger entropy than existing works:
 - a) The different p-bit key values for deciding the number of input resource configurations
 - b) Design parameters chosen for dispersion matrix and Eigen matrix
 - c) Encrypted signature strength (d)
 - d) Size of the encryption key (k)
 - e) Maximum possible value (' AM ', ' MM ') of the resources used, such as adders, multipliers, etc.

➤ **ATTACK SCENARIOS**

- 1) Security against forgery attack
- 2) Security against brute force attack
- 3) Security against false positive and ghost signature search attack

REFERENCE

- [1] C. Pilato, S. Garg, K. Wu, R. Karri, and F. Regazzoni, “Securing hardware accelerators: A new challenge for high-level synthesis,” *IEEE Embedded Syst. Lett.*, vol. 10, no. 3, pp. 77–80, Sep. 2018, doi: 10.1109/LES.2017.2774800.
- [2] R. Schneiderman, “DSPs evolving in consumer electronics applications [Special Reports],” *IEEE Signal Process. Mag.*, vol. 27, no. 3, pp. 6–10, May 2010, doi: 10.1109/MSP.2010.936031.
- [3] H. R. Mahdiany, A. Hormati, and S. M. Fakhraie, “A hardware accelerator for DSP system design,” in *Proc. ICM, Morocco*, 2001, pp. 141–144.
- [4] T. Yu and Y. Zhu, “A new watermarking method for soft IP protection,” in *Proc. Int. Conf. Consum. Electron., Commun. Netw. (CECNet)*, Xianning, China, Apr. 2011, pp. 3839–3842.
- [5] F. Koushanfar, S. Fazzari, C. McCants, W. Bryson, P. Song, M. Sale, and M. Potkonjak, “Can EDA combat the rise of electronic counterfeiting?” in *Proc. DAC Design Autom. Conf.*, San Francisco, CA, USA, Jun. 2012, pp. 133–138, doi: 10.1145/2228360.2228386.
- [6] D. Zhang, X. Wang, Md. T. Rahman, and M. Tehranipoor, “An on-chip dynamically obfuscated wrapper for protecting supply chain against IP and IC piracies,” *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 26, no. 11, pp. 2456–2469, Nov. 2018, doi: 10.1109/TVLSI.2018.2850807.
- [7] A. Cui, C.-H. Chang, S. Tahar, and A. T. Abdel-Hamid, “A robust FSM watermarking scheme for IP protection of sequential circuit design,” *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 30, no. 5, pp. 678–690, May 2011, doi: 10.1109/TCAD.2010.2098131.
- [8] S. G. Rizzo, F. Bertini, and D. Montesi, “Fine-grain watermarking for intellectual property protection,” *EURASIP J. Inf. Secur.*, vol. 2019, no. 1, p. 10, Dec. 2019, doi: 10.1186/s13635-019-0094-2.
- [9] M. Yasin, J. J. Rajendran, O. Sinanoglu, and R. Karri, “On improving the security of logic locking,” *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 35, no. 9, pp. 1411–1424, Sep. 2016, doi:10.1109/TCAD.2015.2511144.

REFERENCE

- [10] A. Sengupta and M. Rathor, “Facial biometric for securing hardware accelerators,” *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 29, no. 1, pp. 112–123, Jan. 2021, doi: [10.1109/TVLSI.2020.3029245](https://doi.org/10.1109/TVLSI.2020.3029245).
- [11] F. Koushanfar, I. Hong, and M. Potkonjak, “Behavioral synthesis techniques for intellectual property protection,” *ACM Trans. Design Autom. Electron. Syst.*, vol. 10, no. 3, pp. 523–545, Jul. 2005, doi: [10.1145/1080334.1080338](https://doi.org/10.1145/1080334.1080338).
- [12] B. Le Gal and L. Bossuet, “Automatic low-cost IP watermarking technique based on output mark insertions,” *Design Autom. Embedded Syst.*, vol. 16, no. 2, pp. 71–92, Jun. 2012, doi: [10.1007/s10617-012-9085-y](https://doi.org/10.1007/s10617-012-9085-y).
- [13] A. Sengupta and S. Bhadauria, “Exploring low cost optimal watermark for reusable IP cores during high level synthesis,” *IEEE Access*, vol. 4, pp. 2198–2215, 2016, doi: [10.1109/ACCESS.2016.2552058](https://doi.org/10.1109/ACCESS.2016.2552058).
- [14] A. Sengupta and M. Rathor, “IP core steganography for protecting DSP kernels used in CE systems,” *IEEE Trans. Consum. Electron.*, vol. 65, no. 4, pp. 506–515, Nov. 2019, doi: [10.1109/TCE.2019.2944882](https://doi.org/10.1109/TCE.2019.2944882).
- [15] University of California Santa Barbara. *Express Benchmarks*. Accessed: Oct. 2023. [Online]. Available: <https://web.ece.ucsb.edu/EXPRESS/benchmark/>
- [16] E. Castillo, L. Parrilla, A. Garcia, U. Meyer-Baese, G. Botella, and A. Lloris, “Automated signature insertion in combinational logic patterns for HDL IP core protection,” in *Proc. 4th Southern Conf. Program. Log.*, Bariloche, Argentina, Mar. 2008, pp. 183–186.
- [17] A. Sengupta and M. Rathor, “Securing hardware accelerators for CE systems using biometric fingerprinting,” *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 28, no. 9, pp. 1979–1992, Sep. 2020, doi: [10.1109/TVLSI.2020.2999514](https://doi.org/10.1109/TVLSI.2020.2999514).
- [18] F. Koushanfar, “Hardware metering: A survey,” in *Introduction to Hardware Security and Trust*, M. Tehranipoor and C. Wang, Eds. New York, NY, USA: Springer, 2012.

REFERENCE

- [19] J. Zhang, Y. Lin, Y. Lyu, and G. Qu, “A PUF-FSM binding scheme for FPGA IP protection and pay-per-device licensing,” *IEEE Trans. Inf. Forensics Security*, vol. 10, no. 6, pp. 1137–1150, Jun. 2015, doi: [10.1109/TIFS.2015.2400413](https://doi.org/10.1109/TIFS.2015.2400413).
- [20] D. B. Roy, S. Bhasin, I. Nikolić, and D. Mukhopadhyay, “Combining PUF with RLUTs: A two-party pay-per-device IP licensing scheme on FPGAs,” *ACM Trans. Embed. Comput. Syst.*, vol. 18, no. 2, p. 22, Mar. 2019, doi: [10.1145/3301307](https://doi.org/10.1145/3301307).
- [21] B. Halak, M. Zwolinski, and M. S. Mispan, “Overview of PUF-based hardware security solutions for the Internet of Things,” in *Proc. IEEE 59th Int. Midwest Symp. Circuits Syst. (MWSCAS)*, Abu Dhabi, United Arab Emirates, Oct. 2016, pp. 1–4.
- [22] A. Anshul and A. Sengupta, “PSO based exploration of multi-phase encryption based secured image processing filter hardware IP core datapath during high level synthesis,” *Expert Syst. Appl.*, vol. 223, Aug. 2023, Art. no. 119927, doi: [10.1016/j.eswa.2023.119927](https://doi.org/10.1016/j.eswa.2023.119927).
- [23] *15 nm Open Cell Library*. Accessed: May 2023. [Online]. Available: <https://si2.org/open-cell-library/>
- [24] M. Potkonjak, “Methods and systems for the identification of circuits and circuit designs,” USPTO Patent U.S. 7 017 043 B1, 2006.
- [25] T. A. Abdel-Hamid, S. Tahar, and E. M. Aboulhamid, “A public-key watermarking technique for IP designs,” in *Proc. DATE*, Munich, Germany, Mar. 2005, pp. 330–335.
- [26] W. Hu, C.-H. Chang, A. Sengupta, S. Bhunia, R. Kastner, and H. Li, “An overview of hardware security and trust: Threats, countermeasures, and design tools,” *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 40, no. 6, pp. 1010–1038, Jun. 2021, doi: [10.1109/TCAD.2020.3047976](https://doi.org/10.1109/TCAD.2020.3047976).
- [27] A. Sengupta and S. P. Mohanty, *IP Core Protection and Hardware-Assisted Security for Consumer Electronics*. The Institute of Engineering and Technology (IET), 2019.

REFERENCE

- [28] *CAD for Assurance*. Accessed: Oct. 2023. [Online]. Available: <https://cadforassurance.org/tools/ip-ic-protection/faciometric-hardwaresecurity-tool/>
- [29] A. Sengupta, R. Chaurasia, and A. Anshul, “Robust security of hardware accelerators using protein molecular biometric signature and facial biometric encryption key,” *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 31, no. 6, pp. 826–839, Jun. 2023.
- [30] J. Liu, J. Xie, L. Chao, H. Wu, P. Ding, X. Chen, and H. Feng, “Minentropy estimation for semiconductor superlattice true random number generators,” *Sci. Rep.*, vol. 12, no. 1, p. 2948, Feb. 2022.
- [31] H. Badier, C. Pilato, J.-C. L. Lann, P. Coussy, and G. Gogniat, “Opportunistic IP birthmarking using side effects of code transformations on high-level synthesis,” in *Proc. Design, Autom. Test Eur. Conf. Exhib. (DATE)*, Grenoble, France, 2021, pp. 52–55.
- [32] S. Rai, S. Garg, C. Pilato, V. Herdt, E. Moussavi, D. Sisejkovic, R. Karri, R. Drechsler, F. Merchant, and A. Kumar, “Vertical IP protection of the next-generation devices: Quo vadis?” in *Proc. Design, Autom. Test Eur. Conf. Exhib. (DATE)*, Grenoble, France, Feb. 2021, pp. 1905–1914.
- [33] C. Pilato, “High-level methods for hardware IP protections: Solutions, trends, and challenges,” in *Proc. IEEE 15th Dallas Circuit Syst. Conf. (DCAS)*, Dallas, TX, USA, Jun. 2022, pp. 1–2.
- [34] J. Bhandari, “Exploring eFPGA-based redaction for IP protection,” in *Proc. IEEE/ACM Int. Conf. Comput. Aided Design (ICCAD)*, Munich, Germany, 2021, pp. 1–9.
- [35] *NIST Computer Security Resource Center*, Glossary, Accessed: Oct. 2023. [Online]. Available: <https://csrc.nist.gov/glossary/term/entropy#:~:text=A%20measure%20of%20the%20amount,is%20usually%20stated%20in%20bits>

Thank You!